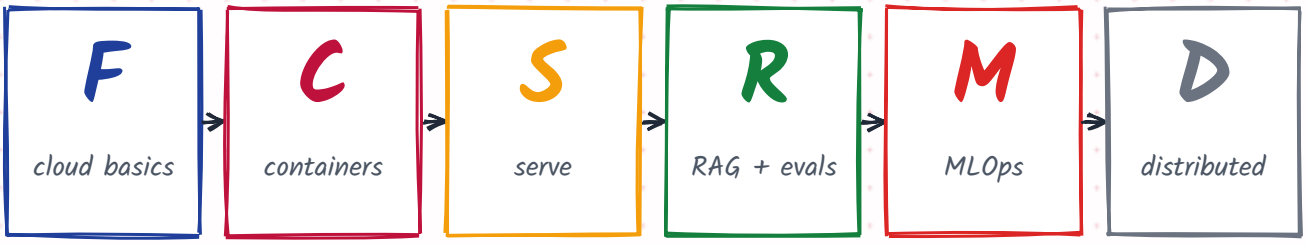
# Your Learning Path

Lesson 12 of 12 · From zero to cloud AI engineer, in 16 weeks.

**In this lesson →** ① Six stops ② Projects and done-when checks ③ Time and money ④ The forgotten GPU ⑤ Certificates in 2026 ⑥ What employers ask ⑦ The claims ledger
**Course project:** ShopNest Copilot, the AI support assistant we moved to the cloud, step by step.

## 1 The road: six stops




**Frugal Coders Spend Really Modest Dollars**

> **F-C-S-R-M-D** = "Frugal Coders Spend Really Modest Dollars": Fundamentals, Containers, Serving, RAG, MLOps, Distributed. Each stop uses the one before. The order is my synthesis (job postings + cost traps), not an official ranking.

- → Run stop 4 (RAG, agents, evals) alongside stops 2-3: almost no cloud spend, and foundation models appear in 51% of postings
- → Every stop ends with a project, an estimated cost and a **done when** check
- → Do not skip the check: being able to show the thing working turns a course into portfolio evidence

## 2 Stops 1 and 2

| Stop | Learn | Project | Done when |
|---|---|---|---|
| **1 Cloud fundamentals (wk 1-2)** | IAM, private network, object storage, GPU quotas, budgets | Terraform: network + least-privilege role + encrypted bucket + budget alert | destroy then apply rebuilds it, no console clicks · other bucket DENIED · budget alert fired |
| **2 Containers + K8s (wk 3-5)** | Dockerfiles for ML, registries, Deployment, autoscaling, GPU node pools | containerised inference service on managed K8s; Terraform GPU pool, one L4, scale to zero | node shows nvidia.com/gpu: 1 · pod runs nvidia-smi · pool back to 0 nodes when idle |

Request GPU quota in week 1: new accounts may need it (unverified for 2026).

## 3 Stops 3 and 4

| Stop | Learn | Project | Done when |
|---|---|---|---|
| **3 Serving (wk 6-7)** | vLLM, SGLang, TGI, Triton, Ray Serve; TTFT, tokens/s, p95 | vLLM serving a 7-8B model on one L4 / A10G / L40S; benchmark | table of tokens/s, TTFT and p95 at 3 concurrency levels x 2 precisions · explain why throughput and p95 both rise |
| **4 RAG, agents, MCP + evals (wk 8-10)** | RAG pipeline, agents, tool calling, MCP, tracing, recall + faithfulness evals | RAG app + eval suite in CI; MCP server + traced agent | CI red when recall drops · every tool call traced · eval set size and gaps stated |

```
# Stage 3 lab: ~7.6B params x 2 B (BF16) = ~15 GB, fits a 24 GB card
vllm serve Qwen/Qwen2.5-7B-Instruct --max-model-len 8192 --gpu-memory-utilization 0.90
for c in 1 8 32; do
  vllm bench serve --model Qwen/Qwen2.5-7B-Instruct --dataset-name random \
    --random-input-len 512 --random-output-len 128 --num-prompts 200 --max-concurrency $c
done
```

## 4 Stops 5 and 6

| Stop | Learn | Project | Done when |
|---|---|---|---|
| **5 MLOps + observability (wk 11-13)** | tracking, registry, CI/CD with eval gate, canary, Terraform environments, drift | LoRA on spot, checkpoints to a bucket, register, serve with vLLM; CI/CD + drift dashboard | kill the spot VM, job resumes, loss curve continuous · versioned model with metrics · alert fires on fake drift |
| **6 Distributed training (wk 14-16)** | DP / TP / PP, ZeRO + FSDP, collectives, MFU and memory accounting | DDP on free 2xT4, layouts on 2-8 rented GPUs, nanochat speedrun (~2 h on 8xH100, about $48, $15 spot) | say from model size whether DDP fits · measured memory roughly agrees · MFU table · chat CLI works |

Spot cost model: spot $ = p_spot x (H + k x h_lost). Without checkpoints, h_lost grows with H (lesson 2).

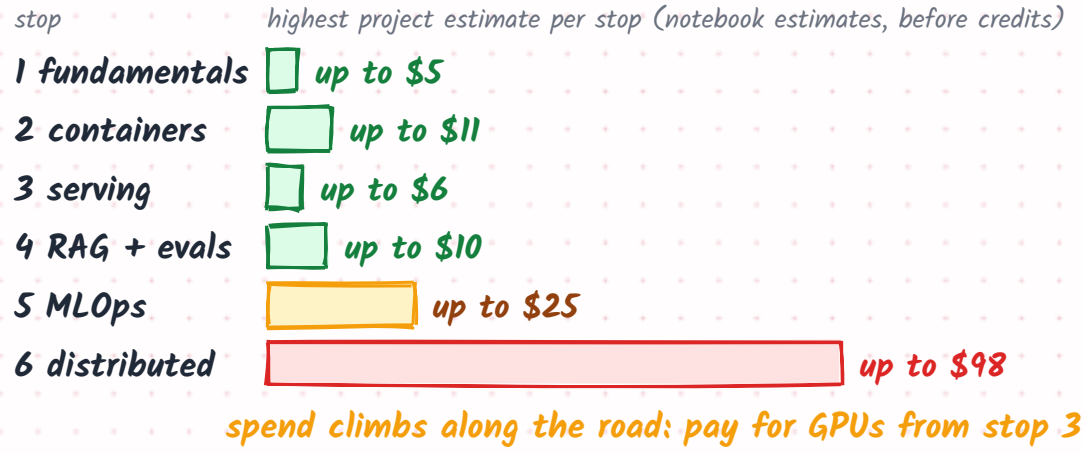
## 5 How long, how much

Arithmetic. 16 weeks x about 10 h = **160 hours** (the 10 is my assumption). Script: l12_study_plan.py.




12-week fast track: skip or merge the four starred weeks (quantisation, agents + MCP, observability, efficiency).

# 6 What the path costs




| | Estimate |
|---|---|
| sum of project estimates (my addition) | $32 to $155 before credits |
| stops 1–5 only | $7 to $57 |
| notebook estimate, whole path | **about $150–250** out of pocket, if you tear down |
| new-account credits (2-3-2) | AWS $200 / 12 months · GCP $300 / 90 days · Azure $200 / 30 days |
| free GPU notebooks | Kaggle about 30 GPU-hours a week |

Budget the higher figure. Spend credits on CPU, networking and Kubernetes stops; use free notebooks for learning-size GPU work; rent marketplace GPUs for stops 3, 5 and 6.

# 7 The forgotten GPU




**one forgotten GPU can cost more than the whole path**

One L4 node left up for a month: 730 h x $0.50-1.20 = **$365 to $876**: 2.4 to 5.7 times the top-end cost of the whole path. Idle resources are the number one way to overspend.

```
# after EVERY lab: prove nothing is still billing
terraform destroy -auto-approve
gcloud container clusters list       # expect: empty
aws eks list-clusters --output text  # expect: empty
```

- ✓ Every Friday: run the teardown commands and check the billing page
- ✓ NAT gateways in "free" networking labs keep billing by the hour: avoid or destroy the same day
- ✓ EKS: $0.10/h standard vs about $0.60/h on extended support: keep versions current
- ✓ GKE $74.40 credit: zonal or Autopilot only, never nodes
- ✓ Spot without checkpoints can cost more than on-demand

## 8 Cheap GPUs by stop

| GPU | About $/GPU-hr (Sep 2026, vendor and aggregator) | Use |
|---|---|---|
| T4 / P100 | free on Kaggle / Colab | stop 1, DDP basics |
| L4 / A10G (24 GB) | $0.50-1.50 | stops 2-3 |
| L40S (48 GB) | $0.92-0.99 | stop 3 benchmarks |
| A100 80 GB | $1.49 (RunPod spot from about $0.36) | stop 5 LoRA |
| H100 | $1.33-3.00 on marketplaces; AWS $6.88 | stop 6 nanochat |

These are vendor and aggregator figures, not provider price lists. No reliable 2026 spot data was found for L4 / A10G.

## 9 Certificates in 2026: check the code

| Retired or changed | What happened | Next |
|---|---|---|
| AWS ML Specialty MLS-C01 | retired, last exam Mar 31, 2026; holders stay active 3 years from date earned | AIP-C01 or MLA-C02 |
| AWS MLA-C01 | English C01 last day Sept 28, 2026 | MLA-C02 (same credential name) |
| Azure AI-102 | retired Jun 30, 2026 | AI-103 (a new exam) |
| Azure DP-100 | retired Jun 1, 2026 | AI-300 MLOps Engineer |
| Azure AI-900 | new version Jun 30, 2026 | AI-901 |

> **Who went MAD**: MLS-C01, AI-102, DP-100. Check the exam code on the official page right before booking.

## 10 Which cert, when

| Stop | Cert (cost) | Note |
|---|---|---|
| 1 | AWS AI Practitioner AIF-C01 ($100) | foundational |
| 2 | CNCF KCNA ($250), then CKA / CKAD ($445) | CKA and CKAD are hands-on, in a live terminal |
| 4 | AWS AIP-C01 ($300) or Azure AI-103 | generative AI applications and agents |
| 5 | Google PMLE ($200) or AWS MLA-C02 or CNCF CNPE ($445) | MLOps, serving, monitoring |
| 6 | NVIDIA NCP-GENL ($200) | LLM professional |

Hands-on exams are probably a stronger signal than multiple choice: inferred from format, not from an employer survey. No data shows how often postings require a specific cert: **projects beat badges**. Prices vary by source: confirm before booking.

## 11 Books and courses: a minimal stack

- → Short on time: Karpathy **Zero to Hero** → **Made With ML** → Huyen **AI Engineering** + HF **Agents** course → **Ultra-Scale Playbook** + **nanochat**
- → Free: fast.ai, HF LLM and MCP courses, Stanford CS336 lectures, Full Stack Deep Learning (2022, tools dated)
- → Huyen **Designing ML Systems** covers classic ML systems; AI Engineering covers foundation-model apps
- → Provider paths (AWS Skill Builder, Google Cloud Skills Boost, Microsoft Learn): the standard free route; catalogs not rechecked

## 12 What employers ask for

| Skill (43,480 US AI-engineering postings to Jul 2026) | Share |
|---|---|
| **Python** | 62% |
| **cloud platforms** | 55% |
| **foundation models** | 51% |

One analyst sample; about 1,550 new US roles a week. Platform / LLMOps roles add vLLM, Kubernetes + GPUs, Terraform, observability and usually want 3 to 7 years. Switching careers: aim at an applied AI engineer role (stops 1-4), add platform skills on the job.

## 13 Name projects in the posting's words

- ✓ "vLLM throughput benchmark"
- ✓ "Terraform GPU node pool"
- ✓ "RAG eval suite in CI"
- ✓ "LoRA on spot with checkpointing"

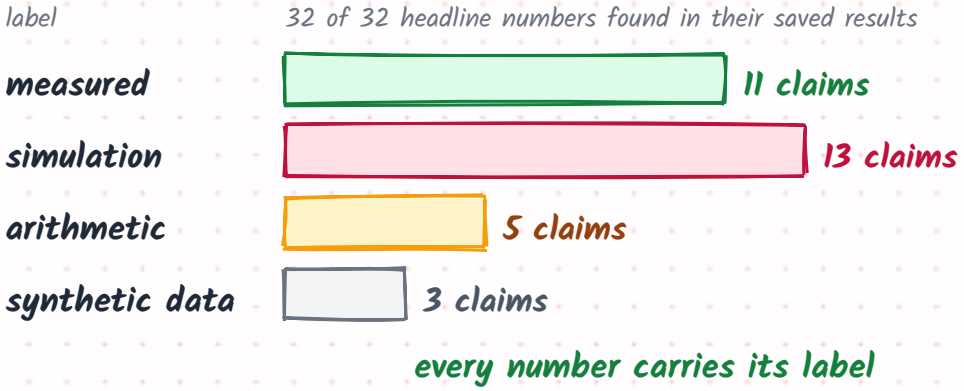
Put those phrases in your repository README files and résumé. Show the number you measured, what it cost, and what it does not cover.

## 14 Double-check before you pay

- → Prices, exam codes and credit rules change monthly: every dollar figure is a dated estimate
- → Sources disagree on several certificate prices and dates
- → Free-trial GPU quota was not rechecked in 2026: do not plan stop 3 around it
- → Posting samples are directional, not guarantees

## 15 The claims ledger

For 32 headline numbers from lessons 1-11, l12_claims_ledger.py looks the number up in the saved output of the lab that produced it and prints its label. **32 of 32 found.**




It proves each claim came from a file you can rerun, not that the experiment is perfect.

## 16 What the labs showed

| Lesson | Headline | Label |
|---|---|---|
| 1 | 70B FP8 + 32 chats of 8,192 tokens + headroom = 187 GB | arithmetic |
| 2 | no checkpoints on spot: 2,203,564 h for a 100 h job | simulation |
| 4 | 20,000 small files: 25x slower than one shard | measured |
| 5 | Adam training state: 16.0 bytes per parameter | measured |
| 6 | KV cache: 12.1x faster generation | measured |
| 7 | eval gate blocked a prompt that raised the average | synthetic |
| 10 | filter 3 of 12 stopped; allowlist 0 leaks | simulation |
| 11 | timestamp on top of a prompt: 0% cached vs 74.9% | measured |

## 17 ShopNest Copilot: the whole plan

| Layer | Decision |
|---|---|
| capacity | fit memory first · spot with checkpoints · scale to zero |
| ship + data | layered images · sharded data · eval gate + canary |
| app | tested chunks · idempotent agent · egress allowlist · tenant filter |
| money | cache first · cost per resolved ticket · budgets and alerts |

## 18 Your first week

- ✓ Open one cloud account and set a budget alert first
- ✓ Request GPU quota now
- ✓ Build the stop 1 Terraform project
- ✓ Run the lab scripts and change one number in each
- ✓ Friday: teardown commands and a billing check

## 19 Words to keep

| Word | Meaning |
|---|---|
| **Teardown** | destroy what you built, then confirm nothing is still billing |
| **Done when** | a check that proves a stage works |
| **Portfolio** | a project you can show running, with its numbers |
| **Label** | is the number measured, simulated, calculated or made up? |

## 20 Interview corner

**Q. Why run RAG and agents alongside the infrastructure stops?**

A. They need almost no cloud spend (about $0-10, mostly API tokens), and foundation models appear in 51% of postings.

**Q. How can a spot run cost more than on-demand?**

A. With no checkpoints, an eviction loses all work: cost is about p_spot x (H + k x h_lost). Checkpoint to object storage and resume.

**Q. How do you avoid a surprise cloud bill?**

A. Budget alert first, tear down after every lab, check for idle resources every week. One forgotten L4 for a month is $365 to $876.

# 21 Interview questions

*"Tell me about a project that shows your cloud AI skills."*

Name it in the posting's words (for example 'vLLM throughput benchmark'), give the number you measured, what it cost, and what it does not cover. Show it running.

**Trap** → Listing tools without a measured result.

*"How would you benchmark an LLM serving setup?"*

vLLM on one L4 / A10G / L40S with a 7-8B model. Record tokens/s, TTFT and p95 at 3 concurrency levels and 2 precisions, and explain why throughput and p95 both rise with concurrency.

**Trap** → Reporting one number at one concurrency level.

*"How do you prove a spot training job is safe?"*

Kill the spot VM on purpose. The job must resume from the last checkpoint in the bucket, with a continuous loss curve. Without checkpoints, spot can cost more than on-demand.

**Trap** → Never testing the resume path until a real eviction happens.

*"How do you keep a learning or dev cloud account cheap?"*

Budget alert first, Terraform so destroy then apply rebuilds everything, teardown after every lab, and a weekly billing check. One forgotten L4 for a month is $365 to $876, more than the whole path.

**Trap** → NAT gateways left running after a 'free' networking lab.

*"Which certificate would you take, and does it matter?"*

One that matches the stop: KCNA then CKA / CKAD for Kubernetes, AWS AIP-C01 or Azure AI-103 for generative AI apps. Check the exam code first: MLS-C01, AI-102 and DP-100 were retired in 2026. Projects beat badges.

**Trap** → Booking a retired exam code.

*"How do you know a number in your portfolio is true?"*

Label every number as measured, simulated, calculated or made up, and link it to a script and its saved output. My claims ledger found 32 of 32 headline numbers in their saved lab results.

**Trap** → Quoting vendor 'up to' figures as if you measured them.

# 22 Course complete

**112_study_plan.py · 112_claims_ledger.py**: change the hours, change a price, rerun.

## ⚡ Revise in 60 seconds

- → 6 stops: F-C-S-R-M-D, run stop 4 alongside
- → 16 weeks at about 10 h a week
- → about **$150–250** if you tear down
- → one forgotten GPU: **$365–876** a month
- → check the exam code before booking
- → projects beat badges
- → name projects in the posting's words
- → label every number you quote

**Course complete**
**Thank you for learning with me.** Go and build your first project this week.

Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**